TEMARL

Section 1 — What changed in the last training run

1–4 are the representation change; 5–6 are the attacker-model change; 7–11 are the training-protocol change.

1	**Step unit**	One technique label → one attacker command. 1,347 labels became 889 commands.
2	**Label handling**	Techniques inside a command became an unordered set.
3	**Artefact removed**	458 of 1,311 transitions (34.9%) had been within-command label order. Now 0%.
4	**Task**	Single-label next-technique → multi-label next-command. Loss changed from cross-entropy to binary cross-entropy over 84 labels.
5	**Markov generator**	Removed entirely. No chain fitting, no smoothing, no synthetic sequences, no generated episodes. Replay only.
6	**New control**	OrderFree added — sees which commands occurred but not their order. Capacity-matched to the other arms.
7	**Window length**	Fixed 16 → tested at 8 / 16 / 32 / 64.
8	**Stopping rule**	Fixed 92-epoch budget → patience 16 / 32 / 64 replayed retrospectively from pilot curves. Chosen: 16 checks.
9	**Training budget**	Measured in optimizer updates, not epochs. Shared budget 4,672 updates, derived from 57 pilot cells.
10	**Convergence**	The previous run carried a cap warning. This run: no configuration unresolved.
11	**Sampler ablation**	50% overlap batching added as an explicitly labelled ablation at W = 16.
12	**Environment**	One command consumes one step and earns at most one unit of reward, however many labels it carries.
13	**Engagement rule**	Set-based: max compatibility over the command's techniques is primary; mean is a separate sensitivity condition.
14	**Pre-registration**	Arms, metrics, families and decision rules frozen in study_spec.py before any confirmatory run.

Deception results — last run

Arm	Dwell	Depth (techniques)	Protection %
Transformer + IPPO	12.769 ± 0.363	13.787 ± 0.351	67.90 ± 4.10
Transformer + MAPPO	13.139 ± 0.193	14.120 ± 0.333	70.00 ± 3.70
GRU + IPPO	12.223 ± 0.217	13.109 ± 0.226	61.30 ± 1.30
GRU + MAPPO	12.469 ± 0.189	13.262 ± 0.183	61.50 ± 1.40
LSTM + IPPO	12.308 ± 0.206	13.192 ± 0.172	61.70 ± 2.20
LSTM + MAPPO	12.702 ± 0.295	13.477 ± 0.293	63.30 ± 2.60
OrderFree + IPPO	12.511 ± 0.443	13.507 ± 0.451	63.70 ± 4.20
OrderFree + MAPPO	12.946 ± 0.393	13.855 ± 0.461	65.50 ± 5.60
NoHistory + IPPO	9.385 ± 0.208	7.429 ± 0.196	58.70 ± 1.40
NoHistory + MAPPO	9.875 ± 0.284	8.011 ± 0.279	63.00 ± 4.60

DESCRIPTION

IDEA 01 — their published method as a baseline on OUR data (prediction half).

IDEA 02 — our model, unchanged, on THEIR released data (prediction half).

IDEA 03 — our method inside their published environment (deception half).

Idea 01 tests - Can their method run on our data?	**Idea 02 tests - Can their data run in our model?**
P1 No invented inputs — every field they read must already exist in our data	**Q1** Actually released with a usable licence ("email us" doesn't count)
P2 Their output maps onto our 84 labels	**Q2** Actions in order, at command or event level
P3 Fully written down — code, or pseudocode + every setting	**Q3** Labels at our granularity, or a published way to derive them
P4 Variance known — deterministic, or we can set the seed	**Q4** Enough data for our train/test split, no leakage

Paper	DOI Link	IDEA 01: Their method on our data	IDEA 02: Our method on their data	IDEA 03: Our method in their environment
HoneyFactory (2024)	https://doi.org/10.3390/electronics13020361	**No, not unchanged:** its HMM predicts attack stages from traffic features; our target is a set of ATT&CK techniques per command.	**No, not directly:** its described traffic datasets do not provide verified command-level ATT&CK labels for our prediction metrics.	**Possible only after rebuilding:** it describes and prototypes a container honeynet, but does not provide a ready-to-run package for our agents; its reported metrics also differ from ours. [Paper]
Yu et al. (2022), Attention Seq2seq	https://doi.org/10.1155/2022/6012232	**Closest prediction comparison, but not unchanged:** it predicts ordered single-technique sequences; our targets are multi-technique command sets. Changing the target format changes the reproduced method.	**Possible for a separate test:** its paper describes public ATT&CK sequence data and a generation tool, but the task and data differ from our command-level evaluation.	**No:** it proposes how prediction could support deception, but does not evaluate a runnable deception controller. [Paper]
ASD-TGXAI (2026)	https://doi.org/10.1016/j.asej.2026.104061	**No, not unchanged:** it combines event sequences and network graphs to classify threats; our task predicts per-command ATT&CK technique sets.	**No, not directly:** its cited Dionaea data are connection records, with no verified ordered per-step ATT&CK labels for our scores.	**Not reproducible as a head-to-head test from the materials found:** the paper reports an adaptive honeypot, but provides no verified runnable package/interface; its reported metrics do not include our full set of dwell, interaction depth, and asset protection. [Paper and data statement]

Section 2

Paper	DOI	Journal / year	Exact model	Dataset / environment	Their metric	Our matching metric	Goal match	Idea 01	Idea 02	Idea 03	Reproducible : what is missing
Kuwano, Okuma, Okada & Mitsunaga [FULL]	https://www.jstage.jst.go.jp/article/ipsjjip/31/0/31_802/_article	Journal of Information Processing 31, 802–811, 2023	User-based collaborative filtering; WkNN with Sørensen–Dice similarity, k=60, support-rate threshold 0.17. Algorithm 1 is complete pseudocode. SupportRate is continuous before thresholding, so ranking metrics are available too.	FULLY PUBLIC. Train = ATT&CK v11 group×technique matrix (133 groups, 191 techniques). Test = CISA AA22-277A (26 techniques), AA22-138B (12 used), AA22-174A (13).	Recall, specificity, accuracy vs number of input techniques. FULL confusion matrices (TP/FP/TN/FN at every input count) in Tables 5–7.	Micro-F1 (primary) and exact-set accuracy. Report their three too — but specificity and accuracy are uninformative at ~4 positives in 84 labels.	STRONG on prediction. Multi-hot observed techniques in, technique set out — exactly our task.	their only input is a technique multi-hot. P2 191 reduce to our 84. P3 Algorithm 1 plus k=60 and threshold 0.17. P4 deterministic.	ATT&CK data and CISA advisories public. Q2/Q3 technique-level. Q4 three independent campaigns as held-out cases.	n/a — no environment.	**NOTHING MISSING. The only paper of 32 reproducible number-for-number: deterministic algorithm, public inputs, exact confusion matrices instead of figures.**
Co-ADAM — Wushishi, Hussain, Hussain et al. [FULL]	https://www.nature.com/articles/s41598-026-60920-0	Scientific Reports 16, 2026	DQN defender (12→128→64→4, 10,180 params) in a constrained MDP with an EMBEDDED SIGNALING GAME: deception investment modulates attacker scepticism, which feeds back into attack success. Co-evolutionary loop vs a tabular attacker; episodes draw from 10 historical attacker	Simulator plus three PUBLIC IIoT datasets: CIC-IIoT-2025, WUSTL-IIoT-2021, Edge-IIoTset. Also a 10-day physical testbed (Dell D660).	AMR 0.8363 (random) / 0.8033 (adaptive). Reward 147.91 / 102.52. Infection 0.0119 / 0.0790. Exploitability 2.4523. Nash gap 0.2137. Convergence episode 1,377 of 5,000. Testbed AMR 0.9975.	Their INFECTION RATE is the honest analogue of our asset-protection rate. AMR is NOT dwell and must not be relabelled as such. No dwell or depth metric exists here.	STRONGEST deception-half match: 3 network zones, 5 tactics, a per-step attacker, and a defender that deploys honeypots and decoys.	FAIL — state is device/infection/scepticism scalars, not a technique history.	n/a — a simulator, not a corpus. Its three IIoT datasets are flow benchmarks.	(a) PASS — attacker acts every step over 100 steps, 45 actions = 5 tactics × 3 intensities × 3 ZONES. (b) PARTIAL — 4 composite profiles weighting 6 mechanisms, so profile selection not per-zone placement, and single-agent DQN not MARL. (c) PARTIAL — infection rate bridges.	**SEEDS AND RUN COUNTS — never stated, so AMR 0.8363 is a point estimate of unknown variance. Code IS released and verified: github.com/Usmanjibril09/CoADAM + Zenodo 10.5281/zenodo.19556576 (CC BY 4.0, 3.1 MB). But it is MATLAB R2025b + 2 toolboxes, not Python; the README's run**

Paper	DOI	Journal / year	Exact model	Dataset / environment	Their metric	Our matching metric	Goal match	Idea 01	Idea 02	Idea 03	Reproducible : what is missing
			snapshots with p=0.30.								**commands point at phases/ core/ figures/ testbed/ subfolders that do not exist; licence is MIT in the README but CC BY 4.0 on Zenodo, with no LICENSE file.**
Wang, Lv, Wang et al. — Shell Attack Evolution Dataset [PARTIAL]	https://ieeexplore.ieee.org/document/11360425	IEEE SRDS 2025, pp. 207–217 — conference; cite with HoneyGPT	No learned model — a data-analysis and dataset-release paper.	PUBLIC, CC BY 4.0 (pipeline MIT). sessions 5,365 + 6,658 = 12,023, fields session_id / src_ip / commands (ordered) / command_count / attack_techniques (ordered, consecutive duplicates collapsed). commands 12,923 + 4,236. request_response 1,512 + 1,489 with turn_index. Techniques 22 → 27.	Descriptive: mean session length 12.65 → 5.00 commands; per-period tactic and technique shares.	Our metrics unchanged. The comparable quantity is the H4 CONTRAST within this corpus — never the absolute micro-F1 against CAM-LDS.	Closest DATA match in the field: ordered real commands plus ordered ATT&CK technique labels, the CAM-LDS structure but from real attackers.	n/a — no model to feed.	CONDITIONAL — Q1, Q2 and Q4 pass. Q3 IS THE OPEN ITEM: labels sit on sessions, not commands. Passes only if their released MIT pipeline re-derives per-command labels deterministically. VERIFY FIRST.	n/a — no environment.	**NOTHING MISSING — data CC BY 4.0, pipeline MIT.**
HoneyGPT — Wang, You, Wang et al. [PARTIAL]	https://doi.org/10.1016/j.comnet.2026.112223	Computer Networks 282, 112223, 2026	LLM-driven SSH/Telnet honeypot on Cowrie with structured prompting. Not a	Uses the corpus above — the 1,489-turn request_response/curated split.	Deception metrics: successful response rate, session length, native	Their session length is not our dwell — no placement decision exists in their design.	WEAK on method, STRONG on data provenance. Carried as the JOURNAL	FAIL P1 — the system emits responses and never reads a technique history.	n/a as a paper — its contribution here is the companion corpus.	FAIL (b) — response generation on one honeypot, no placement.	**NOTHING MISSING — code BSD-3-Clause, data CC BY 4.0. CORRECTS our**

Paper	DOI	Journal / year	Exact model	Dataset / environment	Their metric	Our matching metric	Goal match	Idea 01	Idea 02	Idea 03	Reproducible : what is missing
			trainable sequence model.		capability test. CSVs released.		citation for the corpus above.				**original matrix, which recorded no code; that entry was right when written and is now out of date.**
Lin, Mankaev, Sudyana, Lai & Hwang [FULL]	10.1109/TMLCN.2026.3684465	IEEE Trans. Machine Learning in Comms and Networking 4, 760–779, 2026	TABULAR Q-learning, Algorithm 1 published in full. α=1e-4, γ=0.90, ε_0=0.8 → ε_min=0.01, ρ=0.95, plus a complete 4×2×2 grid. Reward: terminal bonus |T|+1, index(t')+1 on advance, +1 to sustain, −5 on disconnect or detection. State = ICS tactic index, or + protocol type and function code.	Conpot ICS honeypot, Docker and native Python. Scripted Metasploit attacks from MITRE Caldera, each interaction mapped to a MITRE ATT&CK for ICS tactic. Baselines Static0 and Static1.	Static honeypots stall at tactic 4 and tactic 7. RL reaches the latest tactic within 700 attack–response cycles and MORE THAN DOUBLES average engagement time; session duration passes 70 s. Small-state beats Static1 near episode 400, nears tactic 11 by 1000.	CONCEPTUALLY our dwell and depth — NOT the same quantities. Their depth is the MAXIMUM tactic index (ordinal ~1–12); ours is a COUNT of unique techniques (0–84). Their engagement is WALL-CLOCK SECONDS; ours is engaged steps. Neither pair converts.	CLOSEST PURPOSE MATCH in the literature — it optimises what we optimise and penalises detection as our burn hazard does.	FAIL P1 — state is a tactic index plus protocol fields, not a technique multi-hot.	n/a — no corpus released.	(a) PASS — the attacker walks a scripted ICS tactic chain that the honeypot's actions reshape. (b) FAIL — MITRE Engage response shaping (Allow / Modify / Block / Delay) on ONE honeypot; no placement, no zones, no MARL. (c) PARTIAL — see the metric column.	**CODE and the ATTACK SCRIPTS. But Algorithm 1 and every hyperparameter ARE published on a Conpot base, so it is REIMPLEMENTABLE — 'no package' is not 'not reproducible'. No seed count: the paper argues tabular Q-learning removes seed sensitivity instead of reporting runs. Request the Caldera scripts and Conpot config from D. Sudyana (didik@gs.ncku.edu.tw).**
DeepOP — Zhang, Xue & Su [PARTIAL]	10.3390/electronics14020257	Electronics 14, 257, 2025	Transformer with a causal window self-attention mechanism, plus	Attack sequences extracted from CTI reports by ontological	Technique-level sequence prediction. A Transformer baseline at accuracy	Micro-F1, if their output is multi-label rather than top-1 — unconfirmed.	MODERATE — technique-level sequence prediction, but IIoT framing.	UNSCORED — P1 likely fails because the ontology branch needs IIoT asset and	UNSCORED — Data Availability not readable. Author contact	n/a.	**UNKNOWN — MDPI 403. PRIORITY 3.**

Paper	DOI	Journal / year	Exact model	Dataset / environment	Their metric	Our matching metric	Goal match	Idea 01	Idea 02	Idea 03	Reproducible : what is missing
			ontological reasoning over CTI. Multi-step APT prediction for IIoT.	reasoning. Release NOT KNOWN.	0.857 / F1 0.861; DeepOP's own figures not verified.			vulnerability facts, but I could not read the paper to confirm.	drafted (zhangsq@zut.edu.cn).		
Q-Cowrie — Var Naseri, Welch & Haseeb [PARTIAL]	10.1007/s10207-026-01221-5	International Journal of Information Security 25, 2026	Q-learning over an MDP selecting Cowrie responses. Single agent.	More than 400,000 real Cowrie sessions. NOT released — email only; request drafted.	Engagement and session-length measures. Values not verified.	Session length as a dwell proxy, in commands not steps.	MODERATE — response-shaping family, SSH rather than ICS.	FAIL P1 — state is a Cowrie session context, not a technique history.	FAIL Q1 and Q3 — not released, and no ATT&CK labels stated. Would re-enter group B if the email yields labelled data.	(a) PASS in principle — real attackers issue command sequences. (b) FAIL — response selection, not placement.	**DATASET (by email) + CODE. Strictly worse than the TMLCN row, which publishes its full algorithm.**